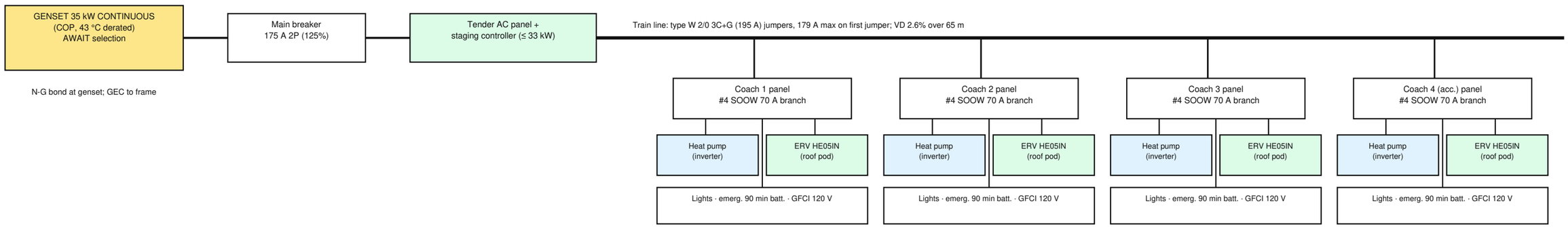
ONE-LINE (120/240 V 1-ph 60 Hz, separately derived, NFPA 70-2023 Art. 445/250/400/522/525)



LOAD AND HVAC BALANCE (calc §8, §10)

Item	Value	Note
Hot-humid demand (plan coach-climate)	39.7 kW	plan basis; 26.8 / 31.7 kW other cases
Outside air, 4 coaches (62.1)	454.0 L/s	279 / 126 cfm per coach; plan 848 L/s kept for G1-b
OA sensible / latent load	10.7 / 8.2 kW	0.4% design day
ERV saving (ε 0.55 ASSUMED)	8.9 kW electric	COP 2.5; ε needed 0.344
Demand WITH ERVs	31.1 / 27.7 kW	plan OA / 62.1 OA; ≤ 34.4 kW target
Staging controller	≤ 33 kW	sheds compressor stages; keeps vent + emergency lights
Rev L4 running load hot / winter	30.94-32.44 / 31.20-32.70 kW	under the 33 kW shed point (Electro E3)
Genset current 35 kW / 240 V	145.8 A	main breaker 175 A
Alternative	45 kW set	needs tender bay re-layout (LT45.1)

NOTES

1. Equipment grounding conductor continuous through every jumper; bonding jumpers across truck pivots.
2. Jumpers: inline locking connectors rated ≥ 200 A, strain relief, interlocked so no live pins exposed.
3. Genset 35 kW continuous (ISO 8528 COP) 1-ph, ≤ 900 kg; tender reservation X1060-2768 (≤ 1,775) × 800 × 1,000 (rev TB-3, basis PTI-38). Cabinet: tender R wall NEMA 3R X2820-3780; listed 200 A 3R UL 1008 ATS w/ integral main; main bkr QOM22225NRB encl. handle ≤ Z1639.
4. ERV RenewAire HE05IN 120 V (880×956×307, 43 kg) in low roof pod ≤ 3,450 ARL; OA/EA louvres separated; MERV 13.
5. Emergency lighting and exit marking battery-backed 90 min per coach (plan EL1).
6. Electrical design to be reviewed/inspected per NFPA 70 and County; wiring by licensed electrician.

PE STAMP / SIGNATURE

(blank - for reviewing NV PE)

UNLESS NOTED: ISO 2768-mK general tolerances
Linear ±0.5 (≤120) / ±0.8 (≤400) / ±1.5 (>400)
Fab. cut ±1.5 · holes pos. ±0.8 · angles ±0.5°
Machined: Ra 3.2 · bores H7/H8 as noted · break edges
GD&T ASME Y14.5-2018 · welds AWS A2.4 / D1.1

PULSIPHER FARMS CHRISTMAS TRAIN — PRELIMINARY, NOT FOR CONSTRUCTION UNTIL PE STAMPED

PULSIPHER FARMS · LOGANDALE, NEVADA · CHRISTMAS TRAIN (PF-CT)

Electrical one-line and coach HVAC (35 kW + ERV)

Sheet E-701 · REV L4 · 2026-10-28 · NTS

Units mm unless noted · Z above top of rail · gauge 914.4 · 3rd-angle

Drawn: owner team 2026-10-27
Checked: _____ Date: ____
Approved (PE): _____